Algebra 1
Unit 1
Lesson 10 Homework

Name Answer Key ____________
Date______________________
Period_____________________

1. To multiply two radicals with the same index, multiply the __coefficients__ together and then multiply the __radicands__ together. Then simplify the radical expression.

For questions 2-7, simplify each radical expression by writing an equivalent radical expression.

2. $\sqrt{13} \cdot \sqrt{13} = \sqrt{169} = \boxed{13}$

3. $\frac{\sqrt{5}}{4\sqrt{125}} = \frac{\sqrt{5}}{4\sqrt{25 \cdot 5}} = \frac{\sqrt{5}}{20\sqrt{5}} = \boxed{\frac{1}{20}}$

4. $\sqrt[3]{6} \cdot \sqrt[3]{9} = \sqrt[3]{54} = \sqrt[3]{27 \cdot 2} = \boxed{3\sqrt[3]{2}}$

5. $\sqrt{24} \cdot \sqrt{8} = \sqrt{192} = \sqrt{64.3} = \boxed{8\sqrt{3}}$

6. $\frac{\sqrt[3]{375}}{\sqrt[3]{648}} = \sqrt[3]{\frac{375}{648}} = \sqrt[3]{\frac{125}{216}} = \boxed{\frac{5}{6}}$

7. $\sqrt[3]{18} \cdot \sqrt[3]{60} = \sqrt[3]{1080} = \sqrt[3]{216 \cdot 5} = \boxed{6\sqrt[3]{5}}$

8. Identify all equivalent expressions to $\sqrt{15} \cdot \sqrt{6}$.

- ○ $9\sqrt{10}$
- ○ $\sqrt{90}$ ✓
- ○ $(10 \cdot 9)^{\frac{1}{2}}$ ✓
- ○ $3\sqrt{15}$
- ○ $3\sqrt{10}$ ✓
- ○ $15^{\frac{1}{3}} \cdot 6^{\frac{1}{2}}$

For questions 9-10, write an equivalent expression by rationalizing the denominator.

9. $\frac{2\sqrt{6}}{\sqrt{5}} \cdot \frac{\sqrt{5}}{\sqrt{5}} = \frac{2\sqrt{30}}{5}$

10. $\frac{\sqrt{28}}{3\sqrt{8}} \cdot \frac{\sqrt{8}}{\sqrt{8}} = \frac{\sqrt{224}}{3 \cdot 8} = \frac{\sqrt{16 \cdot 14}}{24} = \frac{4\sqrt{14}}{24} = \boxed{\frac{\sqrt{14}}{6}}$